Program : **Diploma in Chemical Engineering / Food Processing Technology**	
Course Code : **2072**	Course Title: **Engineering Chemistry**
Semester : **2**	Credits: **3**
Course Category: **Engineering Science**	
Periods per week: **3 (L:2 T:1 P:0)**	Periods per semester: **45**

Course Objectives:

- To orient the students towards chemical engineering as a profession.
- To provide an introduction to basic concepts in inorganic, organic and physical chemistry.
- To demonstrate how the concepts of chemistry help to solve chemical engineering problems.

Course Prerequisites:

Topic	Course Code	Course Name	Semester
Knowledge of basic Mathematics		Mathematics I & II	1, 2
Knowledge of basic Physics		Applied Physics I & II	1, 2
Knowledge of basic chemistry		Applied Chemistry	1

Course Outcomes:

On completion of the course, the student will be able to:

COn	Description	Duration (Hours)	Cognitive level
CO1	Outline the basic concepts of inorganic and analytical chemistry	10	Applying
CO2	Comprehend the fundamental concepts of organic chemistry to solve elementary problems in chemical engineering.	13	Understanding
CO3	Summarise the methods of preparation, chemical properties, reactions and uses of industrially important organic compounds.	13	Understanding

CO4	Apply chemical equilibria and kinetics of chemical reactions for the betterment of chemical engineering.	9	Applying

CO – PO Mapping

Course Outcomes	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7
CO1	2	2					
CO2		3					
CO3	3						
CO4	3	1					

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline

Module Outcomes	Description	Duration (Hours)	Blooms Taxonomy Level
CO1	**Outline the basic concepts of inorganic and analytical chemistry**		
M1.01	Outline the periodicity in properties of elements.	2	Understanding
M1.02	Comprehend analytical chemistry.	3	Understanding
M1.03	Evaluate analytical data	3	Applying
M1.04	Outline the theory behind chromatographic analysis.	2	Understanding
Contents: Periodicity in properties Statement of modern periodic law-explain the Moseley's modern periodic table with emphasis to periods and groups-division of elements in to s, p, d and f block elements, explanation of periodicity of following physical properties along a period and down a group-atomic(covalent and Van der waals radii and ionic radius) .ionization energy, electron affinity and electronegativity. Analytical methods in Chemistry Oxidation, reduction and redox reactions, oxidation number, valency, variable valency, Qualitative analysis – solubility product, common ion effect, application of solubility product and common ion effect in the precipitation of cations during intergroup separation. Evaluation of analytical data, accuracy, precision, errors (absolute and relative error), methods of minimizing error (any three methods), methods of expressing precision- mean, median and standard deviation, simple problems for each.			

<table>
<tr><td colspan="4">Separation by chromatographic techniques
Chromatography- theory- Types of chromatography -Adsorption and partition chromatography, Column chromatography and thin layer chromatography ,Partition chromatography-paper chromatography</td></tr>
<tr><td>CO2</td><td colspan="3">Comprehend the fundamental concepts of organic chemistry to solve elementary problems in chemical engineering.</td></tr>
<tr><td>M2.01</td><td>Comprehend the classification and nomenclature of organic compounds.</td><td>3</td><td>Understanding</td></tr>
<tr><td>M2.02</td><td>Outline the qualitative analysis of organic compounds.</td><td>3</td><td>Understanding</td></tr>
<tr><td>M2.03</td><td>Outline the modern concept of bonding in organic molecules.</td><td>3</td><td>Understanding</td></tr>
<tr><td>M2.04</td><td>Comprehend the types of organic reactions.</td><td>3</td><td>Understanding</td></tr>
<tr><td></td><td>Series Test – I</td><td>1</td><td></td></tr>
<tr><td colspan="4">Contents:
Importance of organic chemistry, general classification of organic compounds , -Homologous series, functional groups, IUPAC nomenclature of organic compounds (Alkanes, alkenes, alkynes, alcohols, aldehydes, ketones, ethers, acids, esters, amines including aromatic compounds).
Qualitative analysis of organic compounds- Give one method of detection of following elements along with the chemistry of reactions - carbon and hydrogen by combustion test , Nitrogen ,sulphur, and halogen by Lassaigne’s test)and phosphorus by ammonium molybdate test .
Modern concept of bonding in organic molecules, SP^3, Sp^2 and SP hybridization in carbon by taking methane, ethane, ethene, benzene and ethyne as examples.
Basic concepts in organic reactions- bond fission, homolytic fission and heterolytic fission, formation and stability of reaction intermediates (carbocations, carbanions and free radicals, 1^0,2^0 and 3^0)
Reagents and reactions in organic chemistry-
Nucleophiles (OH^-,X^-) electrophiles (NO_2^+,SO_3) and free radicals ($Cl\cdot$)
Types of organic reactions-Substitution reactions-
Electrophilic substitution (halogenation, nitration and sulphonation of benzene(no mechanism needed)
Nucleophilic substitution (hydrolysis of methyl halide with aqueous NaOH)
Addition reactions (Addition of Br_2on ethylene)
Elimination reaction (formation of ethylene from 1,2 dibromo ethane in presence of Zn dust)
Rearrangements- formation of isopropyl bromide from n-propyl bromide in presence of $AlBr_3$/HBr</td></tr>
<tr><td>CO3</td><td colspan="3">Summarise the methods of preparation, chemical properties, reactions and uses of industrially important organic compounds.</td></tr>
<tr><td>M3.01</td><td>Outline the preparation and reactions of alkanes.</td><td>3</td><td>Understanding</td></tr>
</table>

M3.02	Outline the preparation and reactions of alkenes.	3	Understanding
M3.03	Outline the preparation and reactions of alkynes.	3	Understanding
M3.04	Outline the preparation and reactions of organic compounds containing oxygen and nitrogen.	4	Understanding

Contents:

Preparation and important reactions of alkanes, alkenes, alkynes and organic compounds containing oxygen and nitrogen

Preparation of hydrocarbons

methane by decarboxylation of carboxylic acid using sodalime distillation-ethane by Wurtz reaction of chloromethane - ethene by dehydrohalogenation of chloroethane using alcoholic KOH – ethyne by action of water on calcium carbide and benzene by cyclic polymerization of ethynes.

Chemical reactions of hydrocarbons –

1. chlorination of methane, ethane and ethyne, Benzene (substitution and addition reactions of chlorine under different conditions)
2. Addition reaction of Br2 in CCl4 to alkenes (test of unsaturation) – Addition of HBr to symmetrical and unsymmetrical alkenes – Markovnikov's and AntiMarkovnikov's addition
3. Acidity of ethyne – reaction with sodium and sodamide (test of alkynes)
4. methylation of benzene (Friedel – Craft's reaction)

Any three uses of methane, ethane, ethene, ethyne and benzene.

Preparation of alcohols and phenol.

ethanol - fermentation reaction and action of Grignard reagent on methanal – reaction of ethanol with ethanoic acid (esterification) – reaction of ethanol with chlorinating agents like Luca's reagent , PCl_3, PCl_5 and $SOCl_2$- oxidation reaction of ethanol using PCC and acidified $K_2Cr_2O_7$ or $KMnO_4$ – three uses of ethanol .

Preparation of phenol – Dow's process (from chlorobenzene) and Cumene process (from benzene)- electrophilic substitution reaction of phenol – bromination (Br_2 in CS_2 and Br_2 in water) and nitration (dil.HNO3 and Conc.HNO3)– special reactions of phenol – Kolbe reaction and Reimer Tiemann reaction (no mechanism is required).

Preparation of aldehydes and ketones

methanal from methanol - ethanal from ethanol – propanone from 2-methyl propan-2-ol, Nucleophilic addition of HCN to methanal and ethanal- Cannizaro's reaction and aldol condensation reaction of methanal and ethanal – oxidation reactions of methanal, ethanal and propanone using strong oxidizing agents like alkaline or acidified $KMnO_4$ and mild oxidizing agents like Tollen's reagent and Fehling's solution (test of aldehydes and ketones).

Structure of glucose – open chain and ring structure- chemical reactions of glucose – reduction using HI, oxidation using Br_2 in water and HNO_3 – acetylation using acetic anhydride and osazone formation using phenyl hydrazine

Preparation of carboxylic acid

Method of preparation of ethanoic acid from CO_2 using Grignard reagent – chemical properties – reduction reaction and HVZ reaction of ethanoic acid .

Preparation of symmetrical and unsymmetrical ethers - Williamson's synthesis (preparation diethyl ether and ethyl methyl ether), formation of epoxides (formation of epoxy ethane from ethylene any three uses of ether.

Preparation of amines

Preparation of aniline- by reduction of nitrobenzene- diazotization of aniline – formation of benzene diazonium chloride – conversion of benzene diazonium chloride to phenol , chlorobenzene (Sandmeyer's reaction and Gattermann reaction) and benzene.
Amino acid –classification of amino acids- essential and nonessential amino acids with two examples each essential amino acids –valine and leucine, nonessential amino acids - glycine and alanine (structures are required) – preparation of glycine by Strecker's synthesis, properties of alpha amino acids- zwitterion structure - definition and significance of isoelectric point – formation of peptide bond and proteins - test of proteins – biuret test ,xanthoproteic and ninhydrin test (no chemistry is required)

CO4	**Apply chemical equilibria and kinetics of chemical reactions for the betterment of chemical engineering.**		
M4.01	Comprehend chemical equilibrium.	2	Understanding
M4.02	Apply laws of chemical equilibrium to solve simple problems.	2	Applying
M4.03	Comprehend chemical kinetics.	2	Understanding
M4.04	Apply chemical kinetics to solve simple problems.	2	Applying
	Series Test – II	1	

Contents:
Chemical Equilibrium: Reversible and Irreversible reactions with two examples for each- Statement of Law of mass action-Equilibrium constants-Equilibrium concentration of reactants and products molar concentration and partial pressure-Derivation of expressions for equilibrium constants-Kc and Kp-Law of chemical equilibrium-Derivation of relation between Kc and Kp-Application of the relation to three chemical equilibria $H_2 + I_2 \leftrightarrow 2HI$, $N_2 + 3H_2 \leftrightarrow 2NH_3$ and $N_2O_4 \leftrightarrow 2NO_2$ and simple problems using this relation Lechateleir's principle-Statement and application of Le – Chateleir's principle to synthesis of ammonia by Haber process and manufacture of sulphuric acid by contact process.
Chemical Kinetics Rate of chemical reaction- definition and unit-Factors affecting rate of chemical reaction -rate law and rate constant-order and molecularity –examples of first order, second order and third order reactions examples of Unimolecular,bimolecular and trimolecular reactions-Difference between order and molecularity of reaction- Derivation of an expression of rate constant of first order reactions-Definition of half life period and derivation of expression of half life period from expression of first order rate constant problems using these expressions-Effect of temperature on reaction rate-Definition of Threshold energy and Activation energy- Arrhenius equation- Exponential forms and logarithmic forms- Explanation of terms- Arrhenius frequency factor and activation energy.

Text / Reference

T/R	Book Title/Author
T1	Organic Chemistry.Vol.–I; I.L.Funar
R1	Engineering Chemistry; P.C.Jain & MonikaJain
R2	Textbook of physical Chemistry; P.L.Soni,O.P.Dharmatha
R3	Textbook of physical Chemistry; Samuel Glasstone
R4	Elements of Physical Chemistry; Glasstone &Lewis

Online Resources

Sl.No	Website Link
1	https://ncert.nic.in/textbook/pdf/kech103.pdf
2	https://www.news-medical.net/life-sciences/Analytical-Chemistry-Techniques.aspx
3	https://chem.libretexts.org/Courses/Lakehead_University/Analytical_I/3%3A_The_Vocabulary_of_Analytical_Chemistry
4	https://chem.libretexts.org/Bookshelves/General_Chemistry/Book%3A_ChemPRIME_(Moore_et_al.)/10%3A_Solids_Liquids_and_Solutions/10.23%3A_Chromatography
5	https://www.organic-chemistry.org/reactions.htm
6	https://chem.libretexts.org/Bookshelves/Physical_and_Theoretical_Chemistry_Textbook_Maps/Supplemental_Modules_(Physical_and_Theoretical_Chemistry)/Equilibria/Chemical_Equilibria/Principles_of_Chemical_Equilibria/Principles_of_Chemical_Equilibrium
7	https://chemed.chem.purdue.edu/genchem/topicreview/bp/ch22/rate.php